SUBJCTIVES OF CURRENT PAPER

BT201 - Ecology, Biodiversity & Evolution-II

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Importance of Bacteria? OR Role of Bacteria in Environment?

The ecosystem, both on land and in the water, depends heavily upon the activity of bacteria. The cycling of nutrients such as carbon, nitrogen, and sulfur is completed by their ceaseless labor. Decomposition is the breakdown of these organisms, and the release of nutrients back into the environment, and is one of the most important roles of the bacteria.

Food Web? / Food Web and Structure of Communities

Definition of Food Web: A food web summarizes the feeding relations in a community. It illustrates the complex interactions between different species, where each species is connected to others through predator-prey relationships.

Structure of Community: The structure of a community can be influenced by the feeding activities of keystone species, which control community dynamics. For instance, in a simple food web, plants are eaten by grasshoppers and squirrels. Predators like foxes and eagles consume these herbivores, and decomposers like earthworms and bacteria break down dead organisms. Keystone species, such as certain predators, can enhance species diversity by preventing competitive exclusion.

Correct sequencing of Organism DNA protein RNA function's arrange them in order:

In organisms, DNA, RNA, and proteins play crucial roles in various functions. Here is the order in which they function:

1. DNA (Deoxyribonucleic Acid): DNA is the genetic material that carries the instructions for the development, functioning, and reproduction of all living organisms. It serves as a blueprint for the synthesis of RNA and proteins.

2. RNA (Ribonucleic Acid): RNA is a molecule that is transcribed from DNA and plays a vital role in protein synthesis. There are different types of RNA, including messenger RNA (mRNA), transfer RNA (tRNA), and ribosomal RNA (rRNA).

3. Protein: Proteins are large, complex molecules made up of amino acids. They perform a wide range of functions in cells, including catalyzing chemical reactions, providing structural support, transporting molecules, and serving as signaling molecules

In summary, DNA provides the instructions, RNA helps in the synthesis of proteins based on those instructions, and proteins carry out various functions essential for the organism's survival and functioning.

Less correlation between mRNA and protein abundance:

The lack of correlation between mRNA and protein abundances is a phenomenon observed in cells. Control mechanisms at the ribosomes, where mRNA is translated to peptides, contribute to this disparity. Translational control mechanisms enable cells to selectively translate certain

mRNAs while blocking others, influencing protein production. This lack of direct correlation suggests that post-transcriptional processes, such as translational control, play a crucial role in regulating protein abundance independently of mRNA levels.

What is industrial ecology:

The design of the industrial infrastructure such that it consists of a series of interlocking "technological ecosystems" interfacing with global natural ecosystems. Industrial ecology takes the pattern and processes of natural ecosystems as a design for sustainability. It represents a shift in paradigm from conquering nature to becoming nature.

What happen when organisms have limited access to resources ?/ Optimal foraging theory

Optimal foraging theory attempts to model how organisms feed as an optimizing process.

Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favor individuals that are more effective at acquiring energy and nutrients. Many animals select food in a way that appears to maximize the rate at which they capture energy. Plants appear to allocate energy to roots versus shoots in a way that increases their rate of intake of the resources that limit their growth. Plants in environments with abundant nutrients but little light tend to invest more energy in the growth of stems and leaves and less in roots. In environments rich in light but poor in nutrients, plants tend to invest more energy in the growth of roots.

Endosymbiotic theory and origin of eukaryotic evolution?

Endosymbiotic Theory: The endosymbiotic theory suggests that eukaryotic cells originated from the symbiotic relationship between several prokaryotic cells. This prolonged association led to the integration of these cells, forming organelles like mitochondria and plastids, which possess their own genetic material and protein-synthesizing capabilities.

Origin of Eukaryotic Evolution: Eukaryotic evolution involved the incorporation of free-living bacteria into a host cell, establishing a symbiotic relationship. Over time, these bacteria became integral parts of the eukaryotic cell, contributing to its complexity and ability to synthesize proteins through distinct systems.

Which is earliest organisms which introduce oxygen in environment?

The earliest organisms to introduce oxygen into the environment were cyanobacteria. These photosynthetic bacteria, also known as blue-green algae, began producing oxygen through photosynthesis around 2.4 to 2.5 billion years ago during the Great Oxygenation Event (GOE). This event significantly increased the levels of oxygen in Earth's atmosphere, paving the way for the evolution of more complex aerobic life forms.

Keystone specie in detail;

A keystone species is a species that has a disproportionately large effect on community structure relative to its biomass or abundance. Keystone species differ from foundation species in two main ways: they are more likely to belong to higher trophic levels (to be top predators), and they act in more diverse ways than foundation species, which tend to modify their environment. Different sources use different definitions for keystone and foundation species. If your class requires you to know these definitions, make sure to consult your book or teacher for the definition you should use. In some sources, foundation species are viewed as a subcategory of keystone species.

Organelles capable of protein synthesis?

Eukaryotic cells contain organelles capable of protein synthesis. These organelles, such as mitochondria and plastids, possess their own genetic material and protein-synthesizing capabilities. This supports the endosymbiotic theory, which suggests that these organelles were once free-living bacteria that entered into a symbiotic relationship with a host cell. This integration eventually led to the development of complex eukaryotic cells, which have a

combination of protein-synthesizing systems derived from both the host and the symbiotic bacteria.

What are transposons?

Transposable elements (insertion sequences and transposons) can transfer copies of themselves to other DNA molecules (chromosome, plasmid, or viral DNA). Antibiotic resistance genes rapidly spread within and between bacterial populations by transposons carried on F factors called R plasmids.

Factor Affecting Decomposition of Environment?

Decomposition rate is influenced by temperature, moisture, and chemical composition of litter and the environment. The rate of decomposition affects the rate at which nutrients, such as nitrogen and phosphorus, are made available to primary producers. Rates of decomposition in terrestrial ecosystems are higher under warm, moist conditions. The rate of decomposition in terrestrial ecosystems increases with nitrogen content and decreases with the lignin content of litter. The chemical composition of litter and the availability of nutrients in the surrounding environment also influence rates of decomposition in aquatic ecosystems.

Differentiation between Shared Ancestral and Shared Derived Characters

A shared ancestral character originated in an ancestor of a taxon and is present in both the ancestor and its descendants. In contrast, a shared derived character is an evolutionary novelty unique to a particular clade, appearing first in the most recent common ancestor of that clade. Systematics use outgroup comparisons to differentiate between these characteristics, determining which traits are ancestral and which are derived by examining how these traits appear in both the in group and closely related outgroup species.

Advantage/Disadvantages to Being Multicelled / single celled

For something on the order of 1.5 billion years all organisms were single-celled and life seems to have thrived. In fact, single-celled organisms are quite good at what they do but what they do is very limited. Single celled organisms cannot grow very large, because as size increases proportionately less of a cell is exposed to the external environment in relation to its volume and the proportion of surface area decreases. Transferring materials from the exterior to the interior becomes less efficient. Multicell organisms live longer, since cells can be replaced and more offspring can be produced. Cells have increased functional efficiency when they are specialized into organs with specific capabilities.

Population Growth?

Population growth is the increase in the number of individuals in a population. Global human population growth amounts to around 83 million annually, or 1.1% per year. OR In the presence of abundant resources, populations can grow at geometric or exponential rates, Population growth by organisms with nonoverlapping generations can be described by the geometric model of population growth, Population growth that occurs as a continuous process, as in human or bacterial populations, can be described by the exponential model of population growth.

Difference between replication, transcription and translation.

Replication: Replication is the process by which DNA is copied to produce identical DNA molecules. It ensures genetic information is passed from cell to cell and generation to generation.

Transcription: Transcription is the process of converting a DNA sequence into messenger RNA (mRNA). It involves RNA polymerase synthesizing mRNA from the DNA template, carrying the genetic code from the nucleus to the ribosome.

Translation: Translation is the process where ribosomes synthesize proteins using the mRNA sequence. Transfer RNA (tRNA) brings amino acids to the ribosome, matching codons on the mRNA to form a polypeptide chain.

Recombination: The exchange of genetic material between different DNA molecules, creating genetic diversity.

Codons or stop codons or nonsense codons.

The stop codons are UAA, UAG, and UGA. They encode no amino acid. The ribosome pauses and falls off the mRNA. They signal release of the nascent polypeptide from the ribosome because no cognate tRNA has anticodons complementary to these stop signals, allowing a release factor to bind to the ribosome instead.

Plant breeding:

Plant breeding is the art and science of changing the traits of plants in order to produce desired characteristics. Plant breeding can be accomplished through many different techniques ranging from simply selecting plants with desirable characteristics for propagation, to methods that make use of knowledge of genetics and chromosomes, to more complex molecular techniques.

Lyell's mistake in natural selection:

As a part of his general resistance to direction/ substantial change in geology, Lyell maintained there was no real direction of change in the fossil record either. This was greeted with pretty general astonishment, but Lyell claimed the apparent changes were just due to our having bad (misleading) samples…more mammals would soon appear in older formations, etc., as we extend our fossil collections. Within ten years (1840) it was clear that this was completely untenable.

Can Humans Act As Keystone Species Give Example?

Humans have acted as keystone species in communities. People have long manipulated food webs both as a consequence of their own feeding activities and by introducing or deleting species from existing food webs. In addition, many of these manipulations have been focused on keystone species. Hunters in tropical rain forests have been responsible for removing keystone animal species from large areas of the rain forests of Central and South America. Chinese farmers have used ants as keystone predators to control pests in citrus orchards for over 1,700 years.

Explain functional response:

The relationship between food density and animal feeding rate is called the functional response. Functional response shapes are typically categorized as linear (type I), hyperbolic (type II), or sigmoidal (type III). The shape of the functional response is generally one of three types. The forms of photosynthetic response curves and type 2 animal functional responses are remarkably similar.

Differentiate between Ecology, Ecological Engineering and Industrial Ecology?

ECOLOGY: Ecology is the scientific study of the processes regulating the distribution and abundance of organisms and the interactions among them, and the study of how these organisms in turn mediate the transport and transformation of energy and matter in the biosphere (i.e., the study of the design of ecosystem structure and function).

INDUSTRIAL ECOLOGY: The design of the industrial infrastructure such that it consists of a series of interlocking "technological ecosystems" interfacin Industrial ecology takes the pattern and processes of natural ecosystems as a design for sustainability. It represents a shift in model from conquering nature to becoming nature.

ECOLOGICAL ENGINEERING: Unlike industrial ecology, the focus of Ecological Engineering is on the manipulation of natural ecosystems by humans for our purposes, using small amounts of supplemental energy to control systems in which the main energy drives are still coming from non-human sources. It is the design of new ecosystems for human purposes, using the self-organizing principles of natural ecosystems.

Effects of Habitat Destruction and Climatic

Warming Long-term studies of populations of land snails around Basel, Switzerland, have documented local extinctions of these land snails. These extinctions are attributable to habitat destruction and climatic warming. The results of these studies suggest that climatic warming can lead to the local extinction of species. As we face the prospect of climatic warming at a global scale, studies of temperature relations will assume greater importance.

Three main sources of energy:

Organisms use one of three main sources of energy: 1= light, 2= inorganic molecules, or 3= organic molecules.

Define Endosymbiont theory. Which organisms introduce the oxygen in atmosphere.

Endosymbiont Theory

The endosymbiont theory posits that eukaryotic cells originated through a symbiotic relationship between multiple prokaryotic cells. Key organelles like mitochondria and plastids, which have their own genetic material and protein synthesis capabilities, were once free-living bacteria that became integrated into a host cell, leading to the development of complex eukaryotic cells.

Oxygen Introduction

Cyanobacteria, also known as blue-green algae, were the earliest organisms to introduce oxygen into the atmosphere through photosynthesis, around 2.4 to 2.5 billion years ago.

Logistic Population Growth

As resources are depleted, population growth rate slows and eventually stops; this is known as logistic population growth. As population size increases, population growth eventually slows and then ceases, producing a sigmoidal, or S-shaped, population growth curve. Population growth stops when populations reach a maximum size called the carrying capacity, the number of individuals of a particular population that the environment can support.

What are distinguishing characteristic of Fungus like protists. Name two group which have mode reproduction:

Fungus-like protists are characterized by their method of reproduction and their appearance. They include slime molds, molds, and mildews, which are often fuzzy growths found on dead material, such as bathroom tiles. Some, like mildew and water molds, are disease-causing, with historical significance such as the potato famine in Ireland.

Groups and Modes of Reproduction

1. **Slime Molds**: Reproduce through spore formation.
2. **Water Molds and Mildew**: Reproduce through both sexual and asexual spores.

What Is the Difference Between Clade, Cladistic, Monophyletic and Paraphyletic?

Cladistics groups organisms by common descent. A clade is a group of species that includes an ancestral species and all its descendants A valid clade is monophyletic, signifying that it consists of the ancestor species and all its descendants A paraphyletic grouping consists of an ancestral species and some, but not all, of the Descendants.

Transduction, transformation and congugation/ mode of gene transfer in Bacteria

1= Transformation: In transformation, a bacterium takes piece of DNA floating in its environment. 2= Transduction: In transduction, DNA is accidentally moved from one bacterium to another by a virus. 3= Conjugation: In conjugation, DNA is transferred between bacteria through a tube between cells.

Explain with Example How Exotic Predators Disturb the Structure Of Food Web.

Exotic predators can collapse and simplify food web structures by devastating native species. For instance, introduced fishes have harmed native fish populations in Lake Atitlan and Gatun Lake in Central America.

Example: Nile Perch in Lake Victoria

The introduction of the Nile perch in Lake Victoria has drastically altered the ecosystem, reducing the diverse fish fauna to a community dominated by only a few species. This highlights the profound impact exotic predators can have on native food webs.

Foundation Species./ exmaples of foundation species.

A foundation species plays a unique, essential role in creating and defining a community. Often, foundation species act by modifying the environment so that it can support the other organisms that form the community, Foundation species are sometimes referred to as 'Ecosystem Engineers'.

Examples of Foundation Species are Corals, Earthworms, Beavers and many more!

Kelp (brown algae) is a foundation species that forms the basis of the kelp forests off the coast of california. Kelps create environments that allow the survival of other organisms that make up the kelp forest community. The corals of a coral reef are another foundation species.

Lamarckism contribution:

Theory of Evolution: Lamarck is best known for his contributions to the theory of evolution, specifically through Lamarckism.

Inheritance of Acquired Traits: Lamarckism suggests that organisms acquire or lose traits based on their usage during their lifetime. According to this theory, traits that are frequently used become stronger and more developed, while those that are not used diminish.

Example: For example, Lamarck proposed that giraffes developed long necks because their ancestors stretched their necks to reach high leaves, and this trait was then inherited by their offspring.

What kind of traits breeders use for crops?

1= Improved quality, such as increased nutrition, improved flavor, or greater beauty, 2= Increased yield of the crop, 3= Increased tolerance of environmental pressures (salinity, extreme temperature, drought), 4= Resistance to viruses, fungi and bacteria, 5= Increased tolerance to insect pests, 6= Increased tolerance of herbicides, 8= Longer storage period for the harvested crop.

Organisms k name puchy thy ciliates, protozaon, flagillates

Ciliates: Paramecium, Stentor

Protozoa: Amoeba, Plasmodium

Flagellates: Euglena, Trypanosoma

Bacteria role in oxidation/reduction in soil:

Bacteria play crucial roles in soil ecosystems by facilitating various oxidation and reduction reactions. These reactions are driven by electron transfers essential for life processes. Bacteria contribute to the breakdown of organic matter, cycling nutrients, and maintaining soil health. They mediate oxidation/reduction processes, such as decomposing organic carbon and transforming nitrogen compounds, thereby supporting plant growth and soil fertility. Oxygenic photosynthesis, powered by sunlight, and the resulting oxygen, further enhance these bacterial activities, driving heterotrophic metabolism in the contemporary biosphere.

Shared ancestral and shared derived characters:

Shared Ancestral Characters

Shared ancestral characters (symplesiomorphies) are traits inherited from a common ancestor that multiple descendants retain. For example, the presence of a backbone is a shared ancestral character for vertebrates, inherited from an early vertebrate ancestor.

Shared Derived Characters

Shared derived characters (synapomorphies) are traits that evolved in the most recent common ancestor of a group and are unique to that group. For example, feathers in birds are a shared derived character, as they evolved in the common ancestor of all birds and are not found in their reptilian ancestors.

Both types of characters are crucial for understanding evolutionary relationships and constructing phylogenetic trees.

Nutrient Loss from Ecosystem?

Major pathways in which these nutrients are lost include: soil erosion, leaching and gaseous losses. Leaching is a physical process where nutrients exit terrestrial ecosystem in the downward flow of water through the soil. ... Another loss of nutrients from ecosystems is due to denitrification.

Why freshwater ecosystems are more threatened?

Freshwater ecosystems are more threatened due to several factors: pollution from agricultural runoff, industrial discharges, and urban waste, leading to degraded water quality; habitat destruction from damming, water extraction, and land development; invasive species disrupting native biodiversity; overfishing depleting aquatic populations; and climate change altering water temperatures and flow patterns. These combined pressures reduce biodiversity, disrupt ecological balance, and impair ecosystem services, making freshwater habitats particularly vulnerable compared to terrestrial and marine ecosystems.

What is the cause of following diseases, African sleeping sickness, plasmodium, protzoans(Inflammation of the large intestine • Diarreha and vomiting)

1= African sleeping sickness is caused by Trypanosoma, 2= plasmodium Causes Malaria, 3= protzoans(Inflammation of the large intestine • Diarreha and vomiting) are caused by amoeba.

Regeneration in echinoderms and how it is useful:

Echinoderms, like starfish, possess remarkable regeneration abilities. When threatened, they autotomize limbs, distracting predators with a nutritious souvenir while escaping. Additionally, they can regenerate from fragments, including the disk., ensuring survival and sustaining their populations despite predation.

Photosynthetic Response?

The relationship between photon flux density and plant photosynthetic rate is called Photosynthetic response.

Which trait breeder wants to incorporate in crops.

Breeders aim to incorporate traits like disease resistance, drought tolerance, higher yield, improved nutritional content, pest resistance, and enhanced growth rates to develop resilient and productive crop varieties.

How to Documented Genetic History in Genome?

1= Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history.

2= DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago

3= mtDNA evolves rapidly and can be used to explore recent evolutionary events.

Nutrients influences by humans aquatic and terrestrial ecosystem?

Human activities enrich nutrients, altering aquatic and terrestrial ecosystems. Increased nitrate and phosphorus levels from human disturbances correlate with population density, reducing plant and fungi diversity. Land managers use nutrient loading models to predict and manage these impacts on aquatic ecosystems.

Correct sequence of organisms, DNA, RNA, protein and function?

1. **Organisms** contain
2. **DNA**, which is transcribed into
3. **RNA**.
4. **RNA** is translated into
5. **Proteins**,
6. which perform various **functions** essential for life processes.

This sequence outlines the flow of genetic information from genes to functional molecules.

Nutrition and nutrients cycle??

Nutrition involves the intake and utilization of nutrients by organisms to maintain life processes. The nutrient cycle, also known as biogeochemical cycling, describes the movement and exchange of essential elements like carbon, nitrogen, and phosphorus through ecosystems. These cycles involve processes such as photosynthesis, decomposition, and nitrogen fixation. Plants absorb nutrients from the soil, which are then transferred to herbivores and predators. Decomposers break down dead organisms, returning nutrients to the soil, thus completing the cycle.

What disease is caused by plasmodium and how it is treated?

plasmodium Causes Malaria, and is Spread by mosquito, and Quinine drug used to treat malaria.

what is indicated by branch length of phylogenetic tree? Also differentiate between maximum parsimony and likelihood.

In some trees, the length of a branch can reflect the number of genetic changes that have taken place in a particular DNA sequence in that lineage. In other trees, branch length can represent chronological time, and branching points can be determined from the fossil record.

Maximum parsimony assumes that the tree that requires the fewest evolutionary events (appearances of shared derived characters) is the most likely

The principle of maximum likelihood states that, given certain rules about how DNA changes over time, a tree can be found that reflects the most likely sequence of evolutionary event

Give the general character of Green Algae:

Green algae are photosynthetic organisms with chlorophyll a and b. They store energy as starch, have cell walls made of cellulose, and can be unicellular or multicellular. Found in freshwater and marine environments, they play a key role in oxygen production.

Traits of Green Algae?

1= Most freshwater, 2= Food Source – called Plankton, 3= Some are: Free-living, Colony, Filament.

Write a detail note on phylum Annelida.

Body Structure: Annelids have segmented bodies with a true coelom, a hydrostatic skeleton, and bristles called setae.

Nutrition: They are heterotrophic, consuming organic matter or preying on small organisms.

Locomotion: Movement is achieved through coordinated contractions of circular and longitudinal muscles, aided by setae.

Reproduction: Annelids reproduce sexually, with many being hermaphroditic. Some can also regenerate lost segments.

Habitat: They inhabit diverse environments, including marine, freshwater, and terrestrial habitats.

Examples: Earthworms, leeches, and polychaetes.

Examples of symbiosis

- **Mutualism**: Bees pollinating flowers, both benefit.
- **Commensalism**: Barnacles on whales, barnacles benefit; whales unaffected.
- **Parasitism**: Ticks on dogs, ticks benefit; dogs harmed.
- **Endosymbiosis**: Mitochondria in eukaryotic cells, both benefit.

Mode of feeding of annelid.

Annelids exhibit diverse feeding modes, including filter feeding, deposit feeding, herbivores, scavengers, and predation. They consume organic matter, detritus, and small organisms. Specialized structures like parapodia and setae aid in locomotion and feeding, while their digestive systems process various food sources efficiently.

Importance of biotechnology:

Biotechnology is crucial for advancing medicine, agriculture, and environmental management. It enables the development of genetically modified crops with enhanced yields, pest resistance, and nutritional value. In medicine, it facilitates the creation of vaccines, gene therapies, and personalized treatments. Environmental applications include bioremediation and sustainable biofuel production. Biotechnology also drives innovations in industrial processes, improving efficiency and reducing waste. Overall, it plays a vital role in addressing global challenges related to health, food security, and sustainability.

Factor Affecting Decomposition of Environment?

Decomposition is influenced by temperature, moisture, oxygen levels, and the chemical composition of organic matter. Microbial activity, presence of decomposers like fungi and bacteria, and soil pH also significantly impact the rate of decomposition.

How does earth depend on redox reaction:

Redox reactions are vital for Earth's biochemical cycles and energy flow. Photosynthesis and respiration are fundamental redox processes that convert sunlight into chemical energy and recycle oxygen and carbon dioxide. In ecosystems, redox reactions facilitate nutrient cycling, breaking down organic matter and releasing essential elements. These reactions also drive energy production in cellular metabolism, supporting all life forms. Additionally, redox reactions play a crucial role in environmental processes, such as detoxification and maintaining soil and water quality.

Importance of biodiversity:

Biodiversity holds immense importance due to its multifaceted functions. It maintains ecosystem balance by recycling nutrients, combating pollution, and stabilizing climate. Intrinsic value underscores its philosophical significance beyond utility. Biodiversity provides crucial biological resources, including medicines like Quinone from plants, essential for human health. Socially, it enriches recreation, tourism, and cultural experiences. Moreover, biodiversity sustains food diversity, crucial for global nutrition, and influences human health through drug discovery. Industries rely on biodiversity for various materials, while it also enriches cultural activities, inspiring art and leisure pursuits.

Write a note on history of biodiversity and some it's milestone.

The history of biodiversity spans over 540 million years, witnessing significant milestones like the Cambrian explosion, where numerous multicellular phyla emerged. Vertebrate diversity displayed an exponential trend over 400 million years, punctuated by mass extinction events, notably the Permian-Triassic extinction. Despite uncertainties about fossil record biases, modern biodiversity estimates range from 2 to 100 million species, with arthropods predominating. Diversity appears to persist and increase over time, even without the influence of natural selection.

How Predators And Parasites Play Role In Population Control?

Predators and parasites have been used to control populations of insects that attack crops or to control invasive weeds. Recent research in Kenya has shown that a crayfish, Procambarus clarkia, controls the snails that act as intermediate hosts for Schistosoma, a highly pathogenic human parasite. Preliminary results indicate that crayfish successfully control host snails in the artificial impoundments used for livestock watering and domestic water, important sources of infection by Schistosoma.

What are vestigial structure?

Vestigial Structures are the Features that have no apparent function. Typically have an important function in related spp. • Ex: Hind limb joints in whales and snakes. • In humans Wisdom teeth, Tail bone, Appendix are vestigial structures.

What are transposons?

Transposable elements (insertion sequences and transposons) can transfer copies of themselves to other DNA molecules (chromosome, plasmid, or viral DNA). Antibiotic resistance genes rapidly spread within and between bacterial populations by transposons carried on F factors called R plasmids.

Characters of Euglenoids and Diatoms?

Euglenoids: Contain chlorophyll, Move by flagella, Euglena (unique), Eyespot, Chloroplast, Flagellum

Diatoms: Float In Water, Photosynthesis, Unicellular, Silica Shells, Commercial value: Insulating materials, Abrasives, Ceramics, Filtering.

Cisgenic and transgenic plants

Cisgenic Plants: Cisgenic plants are genetically modified using genes from the same species or closely related species, preserving genetic integrity while enhancing traits like disease resistance or yield.

Transgenic Plants: Transgenic plants contain genes introduced from unrelated species or organisms, enabling the expression of new traits such as pest resistance, herbicide tolerance, or improved nutritional content. This broader genetic modification can lead to significant advancements in agriculture and crop productivity.

Why breeding of new crops are necessary?

Breeding new crops is necessary to enhance resistance to pests, diseases, and environmental stress, increase yield and nutritional value, and adapt to changing climate conditions. It ensures food security, sustainability, and the ability to meet growing global demands.

Define hibernation with example.

Hibernation is a state of prolonged inactivity and reduced metabolic rate that some animals enter to survive harsh conditions. For example, bears hibernate during winter to conserve energy when food is scarce and temperatures are too low for active survival.

Is share ancestral or shared derived characters occur at same time in a character discuss

Shared ancestral characters (symplesiomorphies) and shared derived characters (synapomorphies) do not occur simultaneously in the evolution of a character. Ancestral characters are inherited from a common ancestor and are present in multiple descendant groups. Derived characters evolve more recently in a lineage, distinguishing more recent common ancestors and their descendants. While ancestral traits are common across broader groups, derived traits are specific to narrower groups, marking evolutionary divergence. Thus, they represent different evolutionary stages and relationships.

Name three effects of organisms:

- **Ecosystem Regulation**: Organisms like wolves control prey populations, maintaining ecosystem balance.
- **Nutrient Cycling**: Decomposers, such as bacteria and fungi, recycle nutrients back into the soil.
- **Pollination**: Bees pollinate plants, facilitating reproduction and food production.

Sigmoidal Population:

Growth Sigmoidal population growth can be modelled by the logistic growth equation, a modification of the exponential growth equation that includes a term for environmental resistance. In the logistic model, the rate of population growth decreases as population density increases. Research on laboratory populations indicates that zero population growth at carrying capacity may be attained by many combinations of reduced birthrates and increased death rates.

Difference Between Genotype and Phenotypes:

Genotype: The genotype refers to the genetic makeup of an organism, including all its alleles and genes inherited from its parents. It determines the potential traits an organism can exhibit.

Phenotype: The phenotype is the observable characteristics or traits of an organism resulting from the interaction between its genotype and the environment. Examples include physical appearance, behavior, and physiological functions. While genotype provides the genetic blueprint, phenotype is the actual manifestation of those genetic instructions.

How lizard kill their prey?

The powerful venom has a devastating effect on prey, causing a sudden drop in blood pressure that sends them into shock. At the same time, the venom stops the victim's blood clotting, so it bleeds to death.

PAR Quantification:

PAR can be quantified as photosynthetic photon flux density, generally reported as µmol per square meter per second Among plants, there are three major alternative photosynthetic pathways, C3, C4, and CAM. C4 and CAM plants are more efficient in their use of water than are C3plants.

Carnivores and herbivore difference:

Carnivores: Animals that primarily eat other animals. They have adaptations for hunting, killing, and consuming meat, such as sharp teeth and claws. Examples include lions and eagles.

Herbivores: Animals that primarily consume plant material. They have adaptations like flat teeth for grinding and specialized digestive systems to process plant matter. Examples include cows and rabbits.

EXTRA MATERIAL (DIL KRY TO AIK BAR PRH LI JIYE GA WRNA OPER WALA MUST KRIYE GA)

Species richness./Specie diversity long 5 marks

Species richness is the number of different species in a particular community. If we found 30 species in one community, and 300 species in another, the second community would have much higher species richness than the first. Communities with the highest species richness tend to be found in areas near the equator, which have lots of solar energy (supporting high primary productivity), warm temperatures, large amounts of rainfall, and little seasonal change. Communities with the lowest species richness lie near the poles, which get less solar energy and are colder, drier, and less amenable to life.

Species diversity is a measure of community complexity. It is a function of both the number of different species in the community (species richness) and their relative abundances (species evenness). Larger numbers of species and more even abundances of species lead to higher species diversity.

For example: A forest community with 20 different kinds of trees would have greater species diversity than a forest community with only 5 kinds of trees (assuming that the tree species were even in abundance in both cases)

Dinoflagellates Why harmful!?

Dinoflagellates pose a threat to aquatic life due to their ability to cause harmful algal blooms, known as red tides. These blooms can lead to the death of fish and other marine animals, as well as human illnesses caused by toxins produced by the dinoflagellates, such as paralytic shellfish poisoning and ciguatera.

What's the difference between a crayfish and a lobster

Same Order, but different families, Lobsters are bigger, Lobsters are marine; crayfish live in freshwater creeks, ditches, or lakes

Barnacles:living and nonliving substrates, most species secrete CaCO3shell, Head reduced, rudimentary abdomen

Krill: Component of plankton, Major food for whales.

How Coral Reefs Act As Foundation Species?

The corals of a coral reef are another foundation species. The exoskeletons of living and dead coral make up most of the reef structure, which protects other species from waves and ocean currents. Beavers, which modify their environment by building dams, can also be seen as a foundation species.

What Intermediate Hypothesis Suggests?

The intermediate disturbance hypothesis suggests that communities with a medium (intermediate) level of disturbance may have greater species diversity than communities with very frequent or very rare disturbances.

Linnaeus Work

Carolus Linnaeus is the father of taxonomy, which is the system of classifying and naming organisms. One of his contributions was the development of a hierarchical system of classification of nature. Today, this system includes eight taxa: domain, kingdom, phylum, class, order, family, genus, and species .

Molds and mildews characters!?!

mold and mildew - disease causing • - fuzzy growths• - found on bathroom tiles• - live off of dead material• - potato famine- 2 million people died in Ireland• - caused immigration to the United States

Genetic Drift

Genetic drift can change gene frequencies in populations, especially in small populations. Genetic drift is theoretically most effective at changing gene frequencies in small populations such as those that inhabit island

Describe the features of exoskeleton of arthropods!?!!

External: not enveloped by living tissue Protection ,Secreted by underlying epidermis

- Waterproof barrier • Chitin +/- calcium, lipoproteins

Modifications • Can be site for muscle attachement • Energy stores- flying • Sensory receptors • Gas exchange

- Bristles

Soft and permeable or hard, impermeable

Between segments of body/appendages= thin + flexible Must be shed (ecdysis= molting) to allow growth Relatively heavy: Limits size

Phylum Echinodermata 5 Classes

Class Crinoidea Class Concentricycloidea Class Stelleroidea Class Echinoidea Class Holothuroidea

Brown algae characters

Kelp (brown algae) is a foundation species that forms the basis of the kelp forests off the coast Of california.

Kelps create environments that allow the survival of other organisms that make up The kelp forest community.

Feeding in leeches

Blood Sucker The salivary glands excrete hirudin which prevents the blood from coagulating. May also secrete an anesthetic and substance to dilate small blood vessels. Blood is broken down by symbiotic bacteria that is then used by the leeches. Leeches were commonly used in the 19th century for bloodletting . Recent medical uses are to relieve pressure after vascular tissue is damaged. Snake bites or the reattachment of a finger or elivin .

why zoos are land of living dead

- They have a very small gene pool in which to mix their genes
- Inbreeding is a serious problem
- Zoos and parks try to solve this by exchanging specimens or by artificial insemination where it is possible
- In vitro fertilisation and fostering by a closely related species has even been tried (Indian Guar – large species of cattle - cloned)
- Even if it is possible to restore a population in captivity the natural habitat may have disappeared in the wild
- Species that rely on this much help are often considered to be “the living dead”

Transduction, transformation and congugation/ mode of gene transfer in Bacteria

Transformation: In transformation, a bacterium takes piece of DNA floating in its environment.

Transduction: In transduction, DNA is accidentally moved from one bacterium to another by a virus.

Conjugation: In conjugation, DNA is transferred between bacteria through a tube between cells.

Landscape Elements

A landscape is a heterogeneous area composed of several ecosystems.

•The ecosystems making up a landscape generally form a mosaic of visually distinctive patches.

•These patches are called landscape elements.

•Landscape ecology is the study of landscape structure and processes.

Landscape ecology?

Landscape ecology is concerned with spatial patterns in the landscape and how they develop, with an emphasis on The role of disturbance, including human impacts‖ (Smith and Smith). It is a relatively new branch of ecology, that Employs Global Information Systems. The goal is to predict the responses of different organisms to changes in Landscape, to ultimately facilitate ecosystem management

Landscape element?

The ecosystems making up a landscape generally form a mosaic of visually distinctive patches. These patches are called landscape elements are the three main source of energy by organism?

How Birds and mammals / Animals regulate body temperature

Birds and mammals rely heavily on metabolic energy to regulate body temperature. The physical nature of the aquatic environment reduces the possibilities for temperature regulation by aquatic organisms.

Most endothermic aquatic species are air breathers. Some organisms, mainly flying insects and some large marine fish, improve performance by selectively heating parts of their anatomy. The energetic requirements of thermoregulation may influence the geographic distribution of species.

Mode of Feeding of Amoeba

Feeding structures- •food vacuole •digestive enzymes •diffusion (ppt)

Amoeba feeds on microscopic organisms such as single-celled algae and bacteria. When the amoeba encounters a suitable organism, the cytoplasm flows round the prey and engulfs it, with a drop of water, in a food vacuole. The cytoplasm secretes enzymes into the food vacuole. (Internet)

Law Of Succession?

General correspondence between fossil and living forms in same geographical area Darwin used this observation to predict that fossils of ancient humans should be found in Africa

Protozoa Life Cycle?

Protozoa have two-phase life cycles, alternating between proliferative stages (e.g., trophozoites) and dormant cysts. As cysts, protozoa can survive harsh conditions, such as exposure to extreme temperatures or harmful chemicals, or long periods without access to nutrients, water, or oxygen for periods of time. (Internet)

Binomial nomenclature...

In the 18th century, Carolus Linnaeus published a system of taxonomy based on resemblances.Two key features of his system remain useful today: two-part names for species and hierarchicalclassification. The two-part scientific name of a species is called a binomial. The first part of thename is the genus. The second part, called the specific epithet, is unique for each species withinthe genus. The first letter of the genus is capitalized, and the entire species name is italicized orunderlined. Both parts together name the species. Binomial nomenclature, also called binominalnomenclature or binary nomenclature, is a formal system of naming species of living things bygiving each a name composed of two parts, both of which use Latin grammatical forms,althoughthey can be based on words from other languages

.**Homology and analogy difference.. 5**

Homology is similarity due to shared ancestry like between a wolf and a coyote

• **Analogy** is similarity due to convergent evolution, similar conditions/adaptations • E.g austrlianmole and northamerican mole lokks alike but but evolved independantly from each other •Analogies are also known as homoplasies (“to mold the same way”)

• **Homology** can be distinguished from analogy by comparing fossil evidence and the degree ofcomplexity• The more complex two similar structures are, the more likely it is that they are homologous **Eusociality**.

Complex level of social behavior, that is considered the pinnacle of social evolution, is called eusociality. Three major characteristics include in Eusociality:

individuals of more than one generation living together, cooperative care of young Division of

individuals into sterile, non-reproductive, and reproductive castes.

Hardy-Weinberg principle,

It states that "in a population mating at random in the absence of evolutionary forces, allele frequencies will remain constant."

Work of Clausen, Keck, and Hiesey

They the sources of morphological variation in plant populations, including both the effect of environment and genetics.

Importance of protistis

Plant-like protists produce almost one-half of the oxygen on the planet through photosynthesis. Other protists decompose and recycle nutrients that humans need to live. ... For example, medicines made from protists are used in treatment of high blood pressure, digestion problems, ulcers, and arthritis.

Why breeding of new crops is important?

Crustaceans characters

• crusta= shell • Lobster, crayfish, shrimp, crab, water flea, barnacles• Aquatic (mostly marine), a few terrestrial forms • Major ecological and economical importance • Biramous appendages (at least primitively), 2 main branches • Only arthropods with 2 pairs of antennae • Great specialization of appendages: Mouthparts chewing, grinding, handling • appendages strengthened for walking or protection (chelipeds, pincer-like claws) • Respiration: gills (usually) • Compound eye is typical of phylum

Eco-line

A gradation from one ecosystem to another when there is no boundary between both systems. It is the joint expression of community and complex environmental gradients.

Ecosystem: Set of organisms and abiotic components connected by the exchange of matter and energy (forest, lake, coastal ocean). Or, "the smallest units that can sustain life in isolation from all but atmospheric surroundings."

Is replication involve in the transformation of RNA form DNA?

No, replication is not involved in the transformation of RNA from DNA. The process of transforming DNA into RNA is called transcription, which occurs in the cell nucleus. Replication, on the other hand, is the process by which DNA makes an identical copy of itself during cell division

Transfer of gene in amoeba

Asexual- create identical organisms

• Conjucation - sexual reproduction

• exchange of genetic information

Not identical - adds diversity